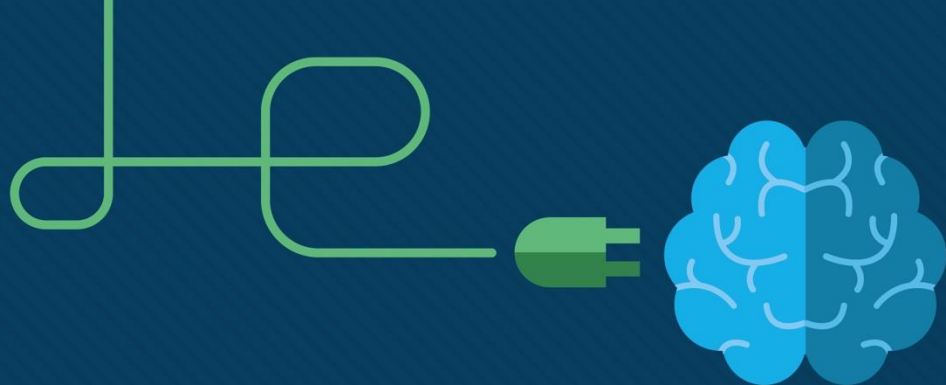




Module 5: STP Concepts

Instructor Materials

Switching, Routing and Wireless
Essentials v7.0 (SRWE)



Module Objectives

Module Title: STP Concepts

Module Objective: Explain how STP enables redundancy in a Layer 2 network.

Topic Title	Topic Objective
Purpose of STP	Explain common problems in a redundant, L2 switched network.
STP Operations	Explain how STP operates in a simple switched network.
Evolution of STP	Explain how Rapid PVST+ operates.

5.1 Purpose of STP

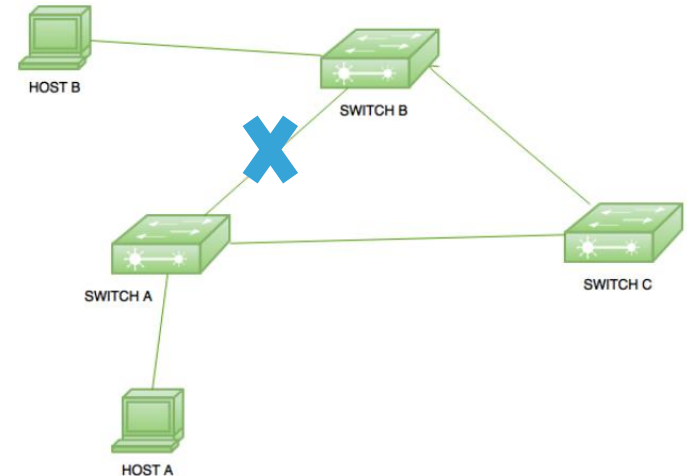
Redundancy in Layer 2 Switched Networks

- **Redundancy** is an important part of the **hierarchical design** for **eliminating single points of failure** and **preventing disruption of network services to users**.
- **Redundant networks** require the addition of **physical** paths.
- Having alternate physical paths for data to traverse the network makes it possible for users to access network resources, despite path disruption. However, **redundant paths** in a switched Ethernet network may cause both **physical and logical Layer 2 loops**.

Role of redundancy is

Avoid single points of failure

Example: if there is a problem in link or connection between switch A and switch B and want to send data from host B to host A



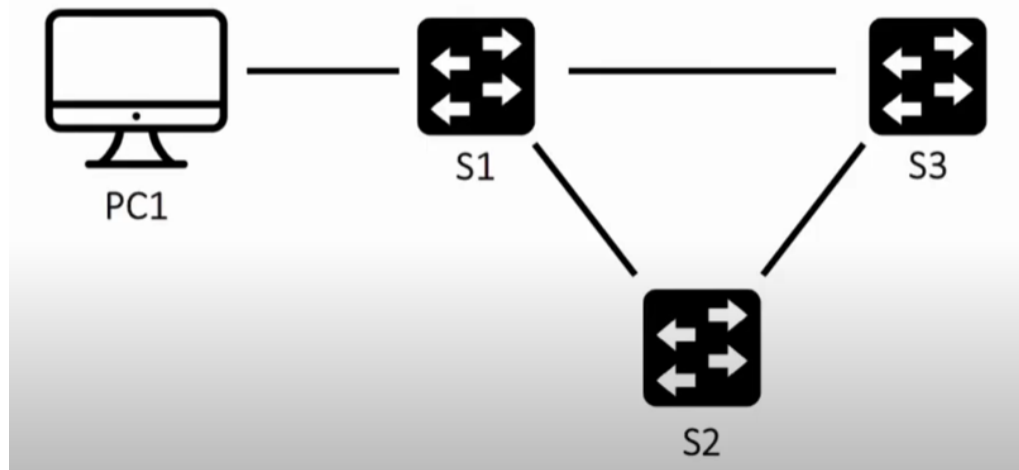
Purpose of STP

Broadcast Storm

- A broadcast storm is an abnormally high number of broadcasts overwhelming the network during a specific amount of time. Broadcast storms can disable a network within seconds by overwhelming switches and end devices. Broadcast storms can be caused by a hardware problem such as a faulty NIC or from a Layer 2 loop in the network.
- Layer 2 broadcasts in a network, such as ARP Requests are very common. Layer 2 multicasts are typically forwarded the same way as a broadcast by the switch. IPv6 packets are never forwarded as a Layer 2 broadcast, ICMPv6 Neighbor Discovery uses Layer 2 multicasts.
- A host caught in a Layer 2 loop is not accessible to other hosts on the network. Additionally, due to the constant changes in its MAC address table, the switch does not know out of which port to forward unicast frames.
- To prevent these issues from occurring in a redundant network, some type of spanning tree must be enabled on the switches. Spanning tree is enabled, by default, on Cisco switches to prevent Layer 2 loops from occurring.

What is looping (broadcast storm)?

What will happen if PC1
send frame broadcasting
to S1?

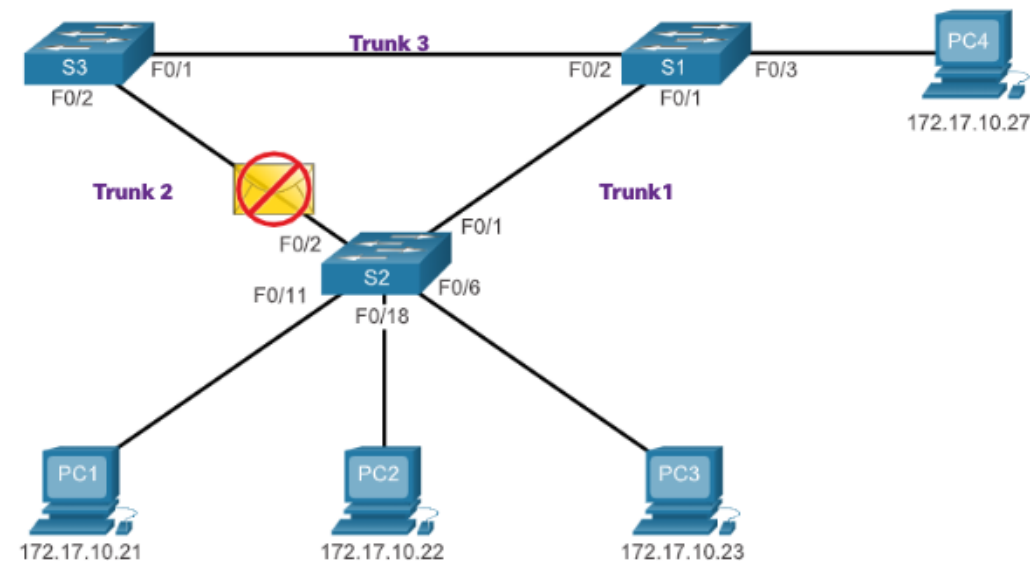


- The data will send from pc1 to sw1 and sw1 will send data to Sw2 and Sw3
 - Sw2 will send broadcast data to all connection except the ingress → to Sw3
 - Sw3 will send broadcast data to all connection except the ingress → to Sw2
 - Sw2 send data to Sw1 also Sw3 will send data to Sw1
- Sw1 will send data to PC1 which discard the data

Purpose of STP

Spanning Tree Protocol STP

- Spanning Tree Protocol (STP) is a **loop-prevention** network protocol that allows for redundancy while **creating a loop-free Layer 2 topology**.
- STP logically **blocks physical** loops in a Layer 2 network, preventing frames from circling the network forever.

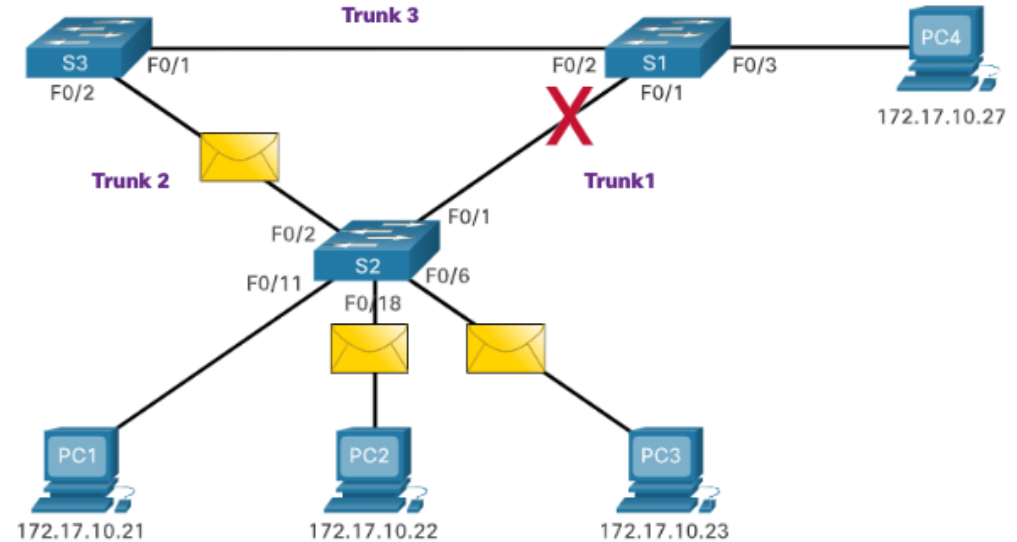


S2 drops the frame because it received it on a blocked port.

Purpose of STP

STP Recalculation (reset the settings)

STP compensates for a failure in the network by recalculating and opening up **previously blocked ports**.



Loop-Free Topology

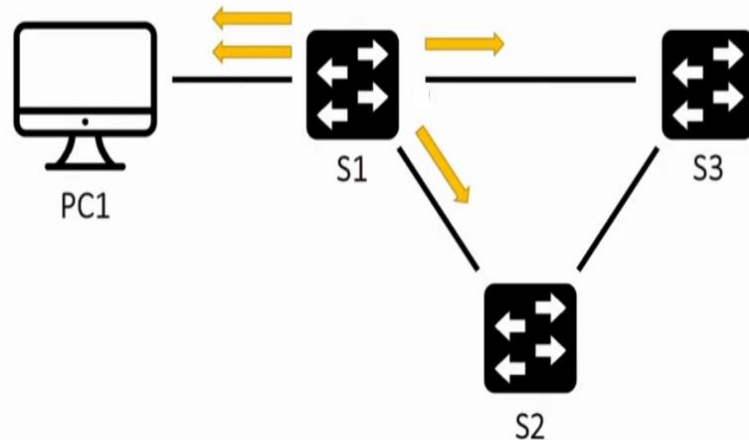
A **loop-free topology** refers to a **network architecture** that is designed with measures to **prevent** the occurrence of network **loops**. Implementing loop-free topologies is crucial for maintaining efficient network performance and **preventing** issues such as broadcast **storms and network paralysis** (stop network).

MAC table instability

The continuous of updating MAC table of switch with incorrect entries, in reaction of looping frame, the result is sending frame to wrong locations.

In the first Sw1 receive frame from pc1 and mac address for pc1.

In the second time Sw1 send data from Sw2 and Sw3 and update the Mac address source for the MAC table.



Purpose of STP

Issues with Redundant Switch Links

- Path redundancy provides multiple network services by eliminating the possibility of a single point of failure.
- When multiple paths exist between two devices on an Ethernet network, and there is no spanning tree implementation on the switches, a Layer 2 loop occurs. A Layer 2 loop can result in MAC address table instability, link saturation, and high CPU utilization on switches and end-devices, resulting in the network becoming unusable.
- **Layer 2 Ethernet does not include a mechanism to recognize and eliminate endlessly looping frames.**
- **Both IPv4 and IPv6 include a mechanism that limits the number of times a Layer 3 networking device can retransmit a packet. A router will decrement the TTL (Time to Live) in every IPv4 packet, and the Hop Limit field (HLF) in every IPv6 packet. When these fields are decremented to 0, a router will drop the packet. Ethernet and Ethernet switches have no comparable mechanism for limiting the number of times a switch retransmits a Layer 2 frame.**
- **STP was developed specifically as a loop prevention mechanism for Layer 2 Ethernet.**

Purpose of STP

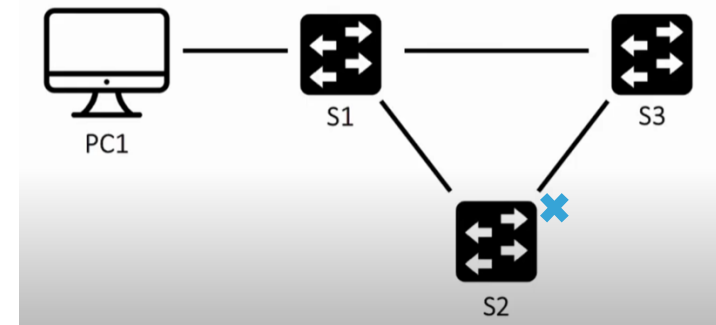
Layer 2 Loops

- Without STP enabled, Layer 2 loops can form, **causing broadcast, multicast and unknown unicast frames to loop endlessly. This can bring down a network quickly.**
- When a loop occurs, the MAC address table on a switch will constantly **change with the updates from the broadcast frames**, which results in **MAC database instability**. This can cause **high CPU utilization**, which makes the switch unable to forward frames.
- **Incorrect data in the MAC address table** gets an unknown unicast frame is when the switch does not have the destination MAC address in its MAC address table and must forward the frame out all ports, except the ingress port.

Purpose of STP

The Spanning Tree Algorithm STA

- STP is based on an algorithm invented by **Radia Perlman** while working for Digital Equipment Corporation, and published in the 1985 paper "An Algorithm for Distributed Computation of a Spanning Tree in an Extended LAN." Her spanning tree algorithm (STA) creates a loop-free topology by selecting a **single root bridge** where all other switches determine a single least-cost path.
- STP prevents loops from occurring by configuring a loop-free path through the network using strategically placed "**blocking-state**" ports. The switches running STP are able to compensate for failures by dynamically unblocking the previously blocked ports and permitting traffic to traverse the alternate paths. → 1. selecting a **single root bridge** 2. **Open the loop of switches by block on port of them.**



The Spanning Tree Algorithm (Cont.)

How does the STA create a loop-free topology?

- **Selecting a Root Bridge:** This bridge (switch) is the reference point for the entire network to build a spanning tree around.
- **Block Redundant Paths:** STP ensures that there is only one logical path between all destinations on the network by intentionally blocking redundant paths that could cause a loop. When a port is blocked, user data is prevented from entering or leaving that port.
- **Create a Loop-Free Topology:** A blocked port has the effect of making that link a non-forwarding link between the two switches. This creates a topology where each switch has only a single path to the root bridge, similar to branches on a tree that connect to the root of the tree.
- **Recalculate in case of Link Failure:** The physical paths still exist to provide redundancy, but these paths are disabled to prevent the loops from occurring. If the path is ever needed to compensate for a network cable or switch failure, STP recalculates the paths and unblocks the necessary ports to allow the redundant path to become active. STP recalculations can also occur any time a new switch or new inter-switch link is added to the network.

5.2 STP Operations

How STP determine specific port to block??

- By using bridge protocol data unit (BPDU).
- =Switches use **Bridge Protocol Data Units (BPDUs)** to share information about themselves and their connections.
- = BPDU is frames contains STP configuration.
- =BPDUs are used to elect the root bridge, root ports, designated ports, and alternate ports.
- BPDU updates every 2 seconds.

Bridge ID (BID)

BPDUs frames contain information of bridge ID.

BID is unique for the switch in the topology.

Bridge ID has information about:

[**priority value** (default 32768 in cisco) + **Extended system ID** (VLAN ID) - **MAC** address switch]



- **Bridge Priority:** The default priority value for all Cisco switches is the decimal value **32768**. The range is **0 to 61440** in increments of **4096**. A lower bridge priority is preferable. A bridge priority of 0 takes precedence over all other bridge priorities.
- **Extended System ID:** The extended system ID value is a decimal value added to the bridge priority value in the BID to **identify the VLAN** for this BPDU.
- **MAC address:** When two switches are configured with the same priority and have the same extended system ID, the switch having the **MAC address with the lowest value, expressed in hexadecimal, will have the lower BID.**

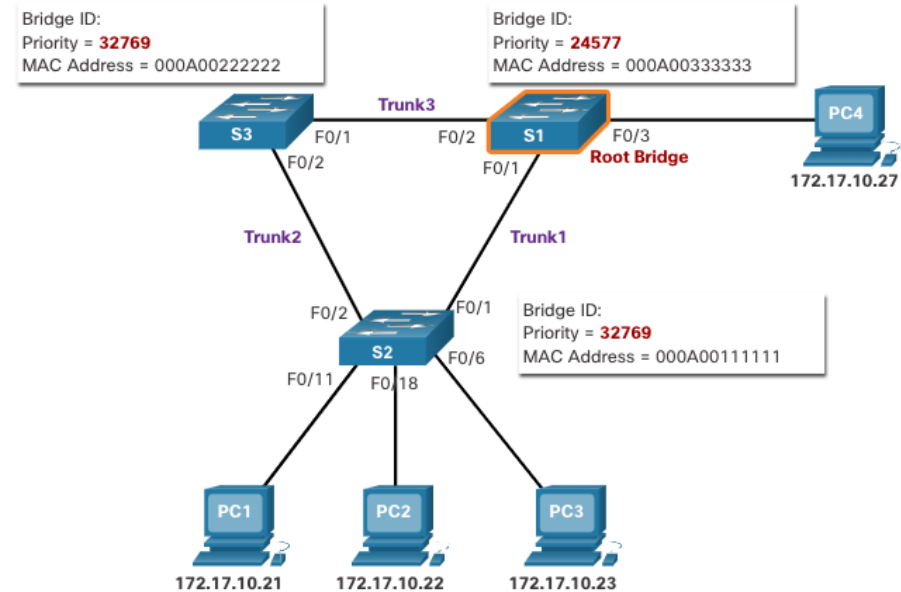
Steps to a Loop-Free Topology

Using the STA, STP builds a loop-free topology in a four-step process:

1. Elect the **root bridge**. (**root switch as reference**)
 2. Elect the **root ports (RP)**. (**transmitting ports**)
 3. Elect **designated ports(DP)**. (**receiving ports**)
 4. Elect **alternate (blocked) ports (Alt)**. (**blocked ports via STP**)
- During STA and STP functions, switches use **Bridge Protocol Data Units (BPDUs)** to share information about themselves and their connections. BPDUs are used to elect the root bridge, root ports, designated ports, and alternate ports.
 - Each BPDU contains a **bridge ID (BID)** that identifies which switch sent the BPDU. The BID is involved in making many of the STA decisions including root bridge and port roles.
 - The BID contains a **priority value**, the **MAC address** of the switch, and an **extended system ID**. The lowest BID value is determined by the combination of these three fields.

1. Elect the Root Bridge (root switch as reference with lower MAC and lower BID)

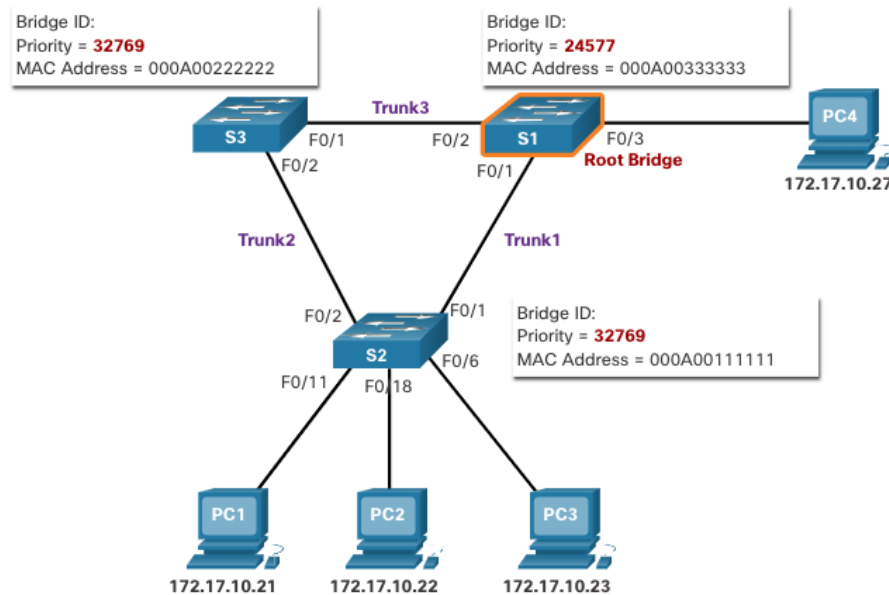
- The STA designates a single switch as the root bridge and uses it as the reference point for all path calculations. Switches exchange BPDUs to build the loop-free topology beginning with selecting the root bridge.
- All switches in the broadcast domain participate in the election process. After a switch boots, it begins to send out BPDU frames every **two seconds**. These BPDU frames contain the **BID** of the sending switch and the BID of the root bridge, known as the Root ID.
- The switch with the **lowest BID** will become the **root bridge**. At first, all switches declare themselves as the root bridge with their own BID set as the Root ID. Eventually, the switches learn through the exchange of BPDUs which switch has the **lowest BID** and will agree on one root bridge.



STP Operations

Impact of Default BIDs

- Because the default BID is **32768**, it is possible for two or more switches to have the same priority. In this scenario, where the priorities are the same, the switch with **the lowest MAC address** will become the root bridge. The administrator should configure the desired root bridge switch with a lower priority.
- In the figure, all switches are configured with the same priority of 32769. Here the MAC address becomes the deciding factor as to which switch becomes the root bridge. The switch with the lowest hexadecimal MAC address value is the preferred root bridge. In this example, **S2** has the lowest value for its **MAC address** and is elected as the root bridge for that spanning tree instance.
- Note:** The priority of all the switches is **32769**. The value is based on the 32768 default bridge priority and the extended system ID (**VLAN 1 assignment**) associated with each switch (**32768+1**).



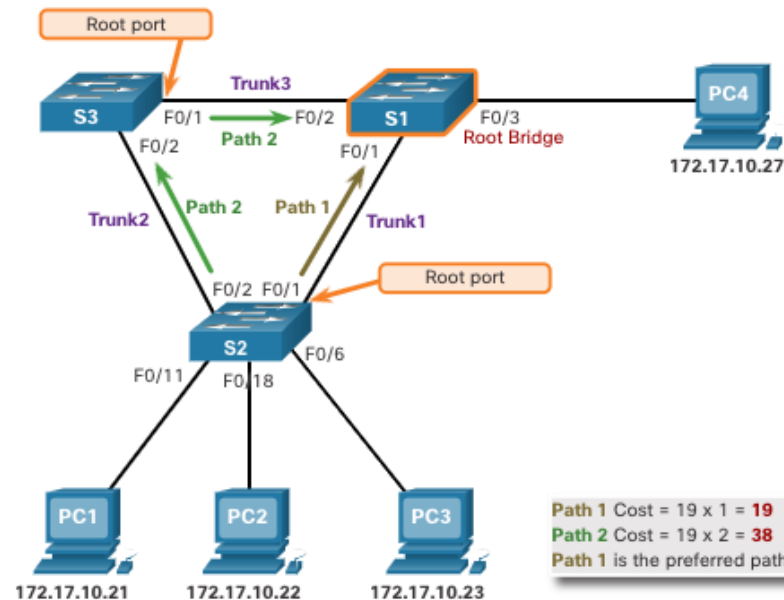
Determine the Root Path Cost (RPC)

- When **the root bridge has been elected** for a given spanning tree instance, the STA starts determining the **best paths** to the **root bridge** from all destinations in the broadcast domain. The path information, known as the internal root path cost, is determined by the **sum of all the individual port costs** along the path from the switch to the root bridge.
- When a switch receives the BPDU, it **adds the ingress port cost** of the segment to determine its internal root path cost.
- The default port costs are defined by the speed at which the port operates. The table shows the default port costs suggested by IEEE. Cisco switches by default use the values as defined by the **IEEE 802.1D** standard, also known as the short path cost, for both STP and RSTP (Rapid Spanning Tree Protocol).
- Although switch ports have a default port cost associated with them, the port cost is configurable. The ability to configure individual port costs gives the administrator the flexibility to manually control the spanning tree paths to the root bridge.

Link Speed	STP Cost: IEEE 802.1D-1998	RSTP Cost: IEEE 802.1w-2004
10 Gbps	2	2,000
1 Gbps	4	20,000
100 Mbps	19	200,000
10 Mbps	100	2,000,000

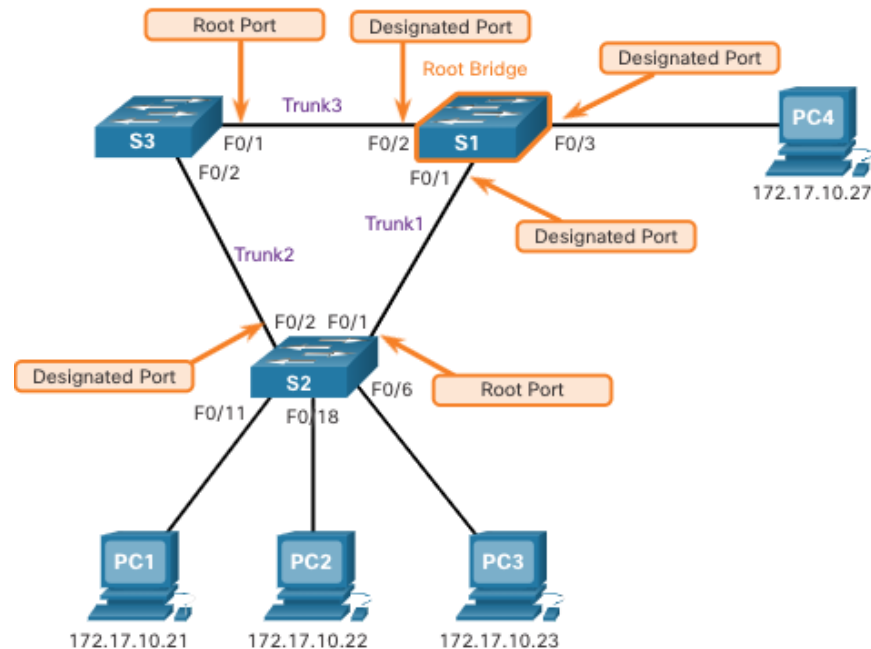
2. Elect the Root Ports (RPs)

- After the root bridge has been determined, the STA algorithm is used to select **the root port**. Every **non-root** switch will select **one root port**. The root port is the port closest to the root bridge in terms of overall cost to the root bridge. This overall cost is known as the internal root path cost.
- The internal root path cost is equal to the sum of all the port costs along the path to the root bridge, as shown in the figure. Paths with the lowest cost become preferred, and all other redundant paths are blocked. In the example, the internal root path cost from S2 to the root bridge S1 over path 1 is 19 while the internal root path cost over path 2 is 38. Because path 1 has a lower overall path cost to the root bridge, it is the preferred path and F0/1 becomes the root port on S2.



3. Elect Designated Ports (DPs)

- Every **segment** between two switches will have **one designated port**. The designated port is a port on the segment that has the internal root path cost to the root bridge. In other words, the designated port has the best path to receive traffic leading to the root bridge.
- What is not a root port or a designated port becomes an alternate or blocked port.
- All ports on the root bridge are designated ports.
- If one end of a segment is a root port, the other end is a designated port.
- All ports attached to end devices are designated ports.
- On segments between two switches where neither of the switches is the root bridge, the port on the switch with the least-cost path to the root bridge is a designated port.



4. Elect Alternate (Blocked) Ports

If a port is **not a root port or a designated port**, then it becomes an **alternate** (or backup) port. Alternate ports are in discarding or blocking state to prevent loops. In the figure, the STA has configured port F0/2 on S3 in the alternate role. Port F0/2 on S3 is in the blocking state and will not forward Ethernet frames. All other inter-switch ports are in forwarding state. This is the loop-prevention part of STP.

